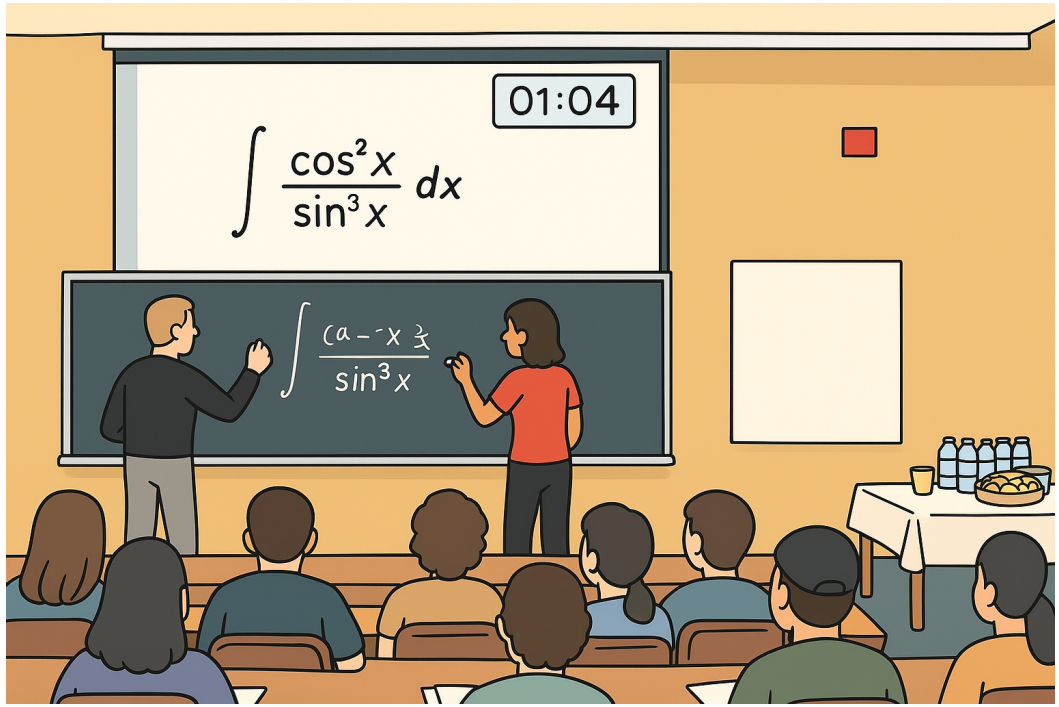
Integratión Bee Physics
Octavos de Final



PRIMER TORNEO DE INTEGRALES
INTEGRATION BEE PHYSICS
CARRERA DE FISICA
UMSA

Octavos de Final

GRUPO 1

Integral #42 **Integratión Bee Physic**

$$\int \frac{e^x \sqrt{e^x - 1}}{e^x + 3} \, \mathrm{d}x$$

Respuesta Integral #42

Integratión Bee Physic

$$\int \frac{e^x \sqrt{e^x - 1}}{e^x + 3} \, \mathrm{d}x =$$

$$2\sqrt{e^x - 1} - 4 \arctan\left(\frac{\sqrt{e^x - 1}}{2}\right) + C$$

Integral #51 **Integratión Bee Physic**

$$\int_0^3 \max\left(x, \sqrt{x+1}\right) \mathrm{d}x$$

Respuesta Integral #51 **Integratión Bee Physic**

$$\int_0^3 \max\left(x, \sqrt{x+1}\right) \mathrm{d}x =$$

$$\frac{5\sqrt{5}+23}{12}$$

Integral #44 Integratión Bee Physic

$$\int \frac{1}{x^3+1}\,\mathrm{d}x$$

Respuesta Integral #44

Integratión Bee Physic

$$\int \frac{1}{x^3+1}\,\mathrm{d}x =$$

$$\frac{1}{3}\ln|x+1| - \frac{1}{6}\ln|x^2-x+1| + \frac{\sqrt{3}}{3}\arctan\left(\frac{2\sqrt{3}x-\sqrt{3}}{3}\right) + C$$

GRUPO 2

Integral #10 **Integratión Bee Physic**

$$\int \frac{1+\tan x}{1-\tan x}\,\mathrm{d}x$$

$$\int \frac{1+\tan x}{1-\tan x}\,\mathrm{d}x =$$

$$-\ln\left|\cos x-\sin x\right|+C$$

Integral #75 Integratión Bee Physic

Sí $$\int \frac{\sqrt{4+x^2}}{x^6}\mathrm{d}x =$$

$$= \frac{A\left(4+x^2\right)^{\frac{3}{2}}\left(Bx^2-6\right)}{x^5} + C$$ ¿Cuánto vale A y B?

$$\int \frac{\sqrt{4+x^2}}{x^6}\mathrm{d}x =$$

$$\frac{1/120\left(4+x^2\right)^{\frac{3}{2}}\left(x^2-6\right)}{x^5} + C$$

$$A = \frac{1}{120} \quad B = 1$$

Integral #12 **Integratión Bee Physic**

$$\int \frac{1}{(1+x^2)\sqrt{1-x^2}}\,\mathrm{d}x$$

Respuesta Integral #12 Integratión Bee Physic

$$\int \frac{1}{(1+x^2)\sqrt{1-x^2}}\,\mathrm{d}x =$$

$$\frac{1}{\sqrt{2}}\tan^{-1}\left[\frac{\sqrt{2}x}{\sqrt{1-x^2}}\right]+C$$

GRUPO 3

Integral #59 **Integratión Bee Physic**

$$\int \frac{\tan x}{\left(\sec^{999} x + 1\right)^2} \mathrm{d}x$$

$$\int \frac{\tan x}{\left(\sec^{999} x + 1\right)^2} \mathrm{d}x =$$

$$\frac{1}{999}\left(\ln\left(\sec^{999} x\right) - \ln\left(\sec^{999} x + 1\right) + \frac{1}{sec^{999} x + 1}\right) + C$$

Integral #87 **Integratión Bee Physic**

$$\int_0^2 x^{x^2+1}(2\ln x + 1)\,\mathrm{d}x$$

Respuesta Integral #87

Integratión Bee Physic

$$\int_0^2 x^{x^2+1}(2\ln x + 1)\,\mathrm{d}x = 15$$

Integral #69 **Integratión Bee Physic**

$$\int \frac{1}{\sin^2 x \cdot \cos^2 x} \mathrm{d}x$$

$$\int \frac{1}{\sin^2 x \cdot \cos^2 x}\, \mathrm{d}x =$$

$$\tan x - \cot x + C$$

GRUPO 4

Integral #32 **Integratión Bee Physic**

$$\int \frac{1}{\sin^2 x \cdot \cos x} \, \mathrm{d}x$$

$$\int \frac{1}{\sin^2 x \cdot \cos x}\,\mathrm{d}x =$$

$$\ln(\sec x + \tan x) + \csc x + C$$

Integral #76 **Integratión Bee Physic**

$$\int \frac{1}{(x-1)^{\frac{3}{4}}(x+2)^{\frac{5}{4}}}\mathrm{d}x$$

Respuesta Integral #76 **Integratión Bee Physic**

$$\int \frac{1}{(x-1)^{\frac{3}{4}}(x+2)^{\frac{5}{4}}}\,\mathrm{d}x =$$

$$\frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4} + C$$

Integral #82 **Integratión Bee Physic**

$$\int_1^2 \left(2^{x-1} + \log_2 2x\right) \mathrm{d}x$$

Respuesta Integral #82 **Integratión Bee Physic**

$$\int_1^2 \left(2^{x-1} + \log_2 2x\right) \mathrm{d}x = 3$$